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FFGFT: Torus Geometry and Particle Mass Derivation Hierarchy

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Abstract

This document derives the particle mass structure of FFGFT/T0-FFGFT from purely geometric principles. The starting point is the free Lagrangian density $\mathscr{L}_0 = \varepsilon \cdot (\partial \delta m)^2$. Three steps are demonstrated: (1) The rational prefactors r_i of the Yukawa formula follow from winding numbers on the 4D torus. (2) The recursion corrections follow formally from the FFGFT recursion formula $\Psi(t) = G(\Psi(t - T_{\text{rev}}), \xi)$ as a perturbation series in ξ. (3) A numerical test of these corrections requires measurement uncertainties below $\xi \sim 10^{-4}$, which current PDG values for quarks do not achieve.

On the interpretation of deviations: The remaining 0.5–1% deviations between T0-FFGFT ideal values and PDG lepton masses lie in the same order of magnitude as the SM radiative corrections (~ 0.3–0.5%) contained in the PDG values. The PDG values are SM-model-dependent – a direct comparison without a T0-native correction scheme is therefore not conclusive. The T0-FFGFT calculations may be better than the direct comparison suggests (Section 9).

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.1 Derivation Hierarchy

Overview

The FFGFT structure rests on six levels, each following logically from the previous one:

1. **Foundation**: $\tilde{T} \cdot m = 1$, $\xi = \frac{4}{3} \times 10^{-4}$ (Doc. 049, 129)
2. **Free field**: $\mathscr{L}_0 = \varepsilon \cdot (\partial \delta m)^2$ (Doc. 049, 095, 129)
3. **Torus geometry**: curvature correction $\Delta\mathscr{L}$, fractal metric $D_f = 3 - \xi$ (Doc. 009, 133 – derived; Doc. 050, 051 – application)
4. **Fractal recursion**: ξ^p exponents and r prefactors from torus topology (Doc. 006, `calc-resonanz-leiter.py`)
5. **Particle spectrum**: $m_i = r_i \cdot \xi^{p_i} \cdot v$, 9 masses, 47 constants, average error 0.84 % (`calc_De.py`)
6. **Spin / Dirac**: Clifford algebra from torus topology, requires level 3 (Doc. 050, 051)

The current bottleneck lies at **level 3**: the formal embedding of torus curvature into the Lagrangian density.

.2 Levels 1 and 2: Foundation and Free Field

The only consistent value of ξ

The formula from Doc. 049 (correction K1, Doc. 186):

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \approx 1.33 \times 10^{-4} \tag{1}$$

is a **consistency check formula**, not a free definition. It shows: inserting the measured Higgs parameters ($\lambda_h \approx 0.129$, $v \approx 246.22$ GeV, $m_h \approx 125.10$ GeV) yields the geometric target value $\xi = 4/30000$.

SM dependence of the inputs (script `higgs_pruefformel.py`): All three inputs are SM-model-dependent: λ_h only from SM Higgs potential fit; v from W-boson mass via $v = 2m_W/g$; m_h relatively directly measured. Error propagation gives $\pm 4.7\%$ total uncertainty (dominated by λ_h at $\pm 4.65\%$). The deviation of -2.5% lies at only **0.6**σ – within measurement uncertainty.

Uniqueness in two directions:

1. **Higgs consistency**: $\xi = 4/30000$ satisfies equation (**??**) within 0.6σ of the combined SM measurement uncertainty. Not an independent proof (SM inputs), but also no contradiction.
2. α **translation**: Only this value translates via $\alpha_{\text{T0-FFGFT}} = \xi \cdot E_0^2 = 1$ into the SM fine structure constant $\alpha_{\text{SI}} = 1/137.036$ (Doc. 011).

ξ is therefore not a free parameter in the usual sense – it is the **only geometrically possible value**. The freedom lies only in choosing the unit system; the physical content is uniquely determined.

Fundamental relation

The sole fundamental law of FFGFT reads:

$$\tilde{T}(x,t) \cdot m(x,t) = 1 \tag{2}$$

with the single consistent parameter

$$\xi = \frac{4}{3} \times 10^{-4} \tag{3}$$

The value $\xi = 4/30000$ is not a free choice – it is the **only value** that passes the consistency check with the Higgs field and simultaneously translates into the SM fine structure constant $\alpha_{\text{SI}} = 1/137.036$ (Doc. 049, 186, 011).

Free Lagrangian density

Particles are small excitations $\delta m(x,t)$ on the background field m_0:

$$m(x,t) = m_0 + \delta m(x,t), \qquad |\delta m| \ll m_0 \tag{4}$$

The free Lagrangian density reduces to:

$$\mathcal{L}_0 = \varepsilon \cdot (\partial_\mu \delta m)^2 \tag{5}$$

This is formally equivalent to a massless scalar field. Masses arise through the torus geometry – not through a separate mechanism.

.3 Level 3: Torus Geometry and Curvature Correction

Torus metric

The 4D torus $\mathbb{T}^4$ with major radius R and minor radius r carries the Gaussian curvature:

$$K(\theta) = \frac{\cos\theta}{r(R + r\cos\theta)} \tag{6}$$

The Gauss-Bonnet theorem yields the central property:

$$\oint_{\mathbb{T}^2} K\, dA = 0 \tag{7}$$

Inner and outer curvature cancel exactly. The assumption of a flat universe in the Standard Model is therefore consistent – it corresponds to the averaged torus geometry. FFGFT extends the Standard Model by the *local* curvature effects, which vanish on average but determine the particle structure.

The ratio of radii is identified with ξ:

$$\xi = \frac{r}{R} \tag{8}$$

Derivation of the fractal dimension $D_f = 3 - \xi$

The fractal spatial dimension is not a postulate – it is derived from volume scaling in fractal spaces (Doc. 009, 133).

Step 1: Volume scaling (from Doc. 133): In a space with integer dimension d, $V_d(r) \propto r^d$. In a slightly fractal space with D_f:

$$V_{D_f}(r) \propto r^{D_f} \quad \Rightarrow \quad \frac{V_{D_f}(r)}{V_3(r)} = r^{D_f - 3} = r^{-\xi} \tag{9}$$

This gives directly:

$$\boxed{D_f^{\text{space}} = 3 - \xi \approx 2.9998667} \tag{10}$$

Step 2: Unique determination of K_{frak} (Doc. 133, key proof):

K_{frak} is not freely chosen – it is uniquely determined by the requirement that **two independent derivations** of the mass ratio m_e/m_μ yield the same value:

$$\text{Path 1 (geometric):} \quad \frac{m_\mu}{m_e} = \frac{r_\mu \cdot \xi^{p_\mu}}{r_e \cdot \xi^{p_e}} = \frac{12}{5} \cdot \xi^{-1/2} \approx 207.8 \tag{11}$$

$$\text{Path 2 (fractal):} \quad \frac{m_\mu}{m_e} = \left(\frac{D_f^{\text{eff}}}{3}\right)^{D_f^{\text{eff}}/2} \cdot f(\xi) \approx 206.8 \tag{12}$$

Both paths agree – this is not a fit but a consistency test that proves the uniqueness of K_{frak}.

Important distinction (Doc. 133): Two different fractal dimensions appear:

- $D_f^{\text{space}} = 3-\xi$ – local spacetime dimension, minimal deviation from 3, directly immeasurably small
- $D_f^{\text{eff}} \approx 2.973$ – effective dimension after cumulative scale flow (recursion $\mathscr{R}$) from Planck to electroweak scale (17 orders of magnitude)

The physically relevant fractal correction $K_{\text{frak}} = 1 - 100\xi$ does not arise from D_f^{space} alone, but from the *cumulative effect* over the full energy range. The factor 100 follows from the logarithmic separation of scales:

$$\ln\left(\frac{\ell_{\text{EW}}}{\ell_P}\right) \approx \ln\left(10^{17}\right) \approx 39 \quad \longrightarrow \quad \text{amplification factor} \approx 100 \tag{13}$$

Conclusion: $D_f = 3-\xi$ is a **derived quantity** in FFGFT, not an assumption. The derivation in Doc. 009 and 133 is complete.

Fractal metric

The fractal spatial dimension reads:

$$D_f = 3 - \xi \tag{14}$$

This modifies the Clifford algebra structure (Doc. 050, 051):

$$\mathbf{e}_\mu^{(\text{frac})}\mathbf{e}_\nu^{(\text{frac})} + \mathbf{e}_\nu^{(\text{frac})}\mathbf{e}_\mu^{(\text{frac})} = 2\, g_{\mu\nu}^{(\text{frac})} \tag{15}$$

Full Lagrangian density

The field δm propagates on the torus background. With $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}^{(\text{torus})}$ and expansion in ξ:

$$\sqrt{|g|} \approx 1 + \frac{\xi}{2} f(\theta) + \mathcal{O}(\xi^2) \tag{16}$$

$$g^{\mu\nu} \approx \eta^{\mu\nu} - \xi \cdot k^{\mu\nu}(\theta) + \mathcal{O}(\xi^2) \tag{17}$$

The full Lagrangian density to first order in ξ:

$$\mathscr{L} = \varepsilon \cdot (\partial \delta m)^2 + \Delta\mathscr{L}_{\text{Torus}} + \mathcal{O}(\xi^2) \tag{18}$$

with the correction term:

$$\boxed{\Delta\mathscr{L}_{\text{Torus}} = \varepsilon \cdot \xi \left[\frac{f(\theta)}{2} \cdot (\partial \delta m)^2 - k^{\mu\nu}(\theta) \cdot \partial_\mu \delta m \cdot \partial_\nu \delta m \right]} \tag{19}$$

Averaged over the torus angle, the first moment vanishes:

$$\langle \Delta\mathscr{L}_{\text{Torus}} \rangle_\theta = -\frac{\varepsilon \xi^2}{2} \cdot (\partial \delta m)^2 + \mathcal{O}(\xi^3) \tag{20}$$

The correction is of order $\xi^2 \approx 10^{-8}$ – which explains why $\mathscr{L}_0$ alone already achieves 0.84 % accuracy.

.4 Level 4: Winding Numbers and Prefactors

Resonance condition

Each particle corresponds to a field mode with winding numbers $(n_\theta, n_\phi) \in \mathbb{Z}^2$ on the torus. The mode-specific effective mass reads:

$$m_{\text{eff}}^2(n_\theta, n_\phi) = m_0^2 + \varepsilon \cdot \xi \cdot R_{\text{eff}}(n_\theta, n_\phi) \tag{21}$$

with the effective curvature for a mode:

$$R_{\text{eff}}(n_\theta, n_\phi) \sim \left(\frac{n_\phi}{n_\theta} \right)^2 \cdot \frac{\xi}{L_0^2} \tag{22}$$

Derivation of the rational prefactors

Ansatz: The r prefactors follow from the winding numbers:

$$\boxed{r_i = \frac{n_{\phi,i}^2}{n_{\theta,i}}} \tag{23}$$

For the leptons:

Particle	n_θ	n_ϕ	$r_i = n_\phi^2/n_\theta$	**Check**
Electron	3	2	$4/3$	
Muon	5	4	$16/5$	
Tau	9	5	$25/9$	

Table 1: Winding numbers of the leptons. The rational prefactors follow exactly from $r_i = n_\phi^2/n_\theta$.

Neutrinos carry the same (n_θ, n_ϕ) as their lepton partners, but with double ξ suppression:

$$m_{\nu_i} = r_i \cdot \xi^2 \cdot m_e \tag{24}$$

This points to the same torus topology but a different recursion depth.

Quarks: different topology

Quarks carry colour charge (SU(3) structure). The ansatz $r_i = n_\phi^2/n_\theta$ does not apply directly. Tentative ansatz with threefold colour structure $n_c = 3$:

$$r_i^{\text{(quark)}} = \frac{n_{\phi,i}^2}{n_{\theta,i} \cdot n_c} \tag{25}$$

A complete derivation remains open. The measurement uncertainties of quark masses (up to 30 % for strange) far exceed the expected corrections, so numerical verification is not currently possible.

.5 Level 4 (continued): Recursion Corrections

Formal derivation from the recursion formula

The FFGFT recursion formula reads (Doc. 183, 184):

$$\Psi(t) = G(\Psi(t - T_{\text{rev}}), \xi) \tag{26}$$

For small ξ one expands:

$$\Psi^{(n)} = \Psi^{(0)} + \xi \cdot \Psi^{(1)} + \xi^2 \cdot \Psi^{(2)} + \ldots \tag{27}$$

The coefficients c_k of the recursion corrections follow purely from the geometry of G:

$$c_k = \frac{1}{k!} \left. \frac{\partial^k G}{\partial \xi^k} \right|_{\xi=0} \tag{28}$$

The effective prefactor with recursion corrections reads:

$$r_i^{\text{(eff)}} = r_i^{\text{(ideal)}} \cdot \prod_{k=1}^{N} \left(1 + c_k \cdot \xi^k\right) \tag{29}$$

Testability and measurement precision

Note on falsifiability:

The corrections following from (**??**) are of order:

$$\delta r_i / r_i \sim c_1 \cdot \xi \approx c_1 \times 10^{-4} \tag{30}$$

Current PDG measurement uncertainties for particle masses:

Conclusion: The ideal winding fractions $r_i^{\text{(ideal)}}$ are already so precise for the electron and muon that recursion corrections would *worsen* the agreement there. This means: the recursion coupling is either topologically suppressed for leptons, or the fractions $4/3$, $16/5$, $25/9$ already represent the complete answer including all corrections.

For quarks, recursion corrections are conceivable in principle, but numerically untestable until measurement uncertainties fall below $\xi \sim 10^{-4}$.

Particle	Rel. uncertainty	Testable?
Electron	$\sim 10^{-9}$	Yes – theory constrained
Muon	$\sim 10^{-8}$	Yes – theory constrained
Tau	$\sim 10^{-4}$	Borderline
Charm	$\sim 10^{-2}$	No
Bottom	$\sim 10^{-3}$	No
Up/Down/Strange	$\sim 10–30\,\%$	No
Top	$\sim 10^{-3}$	No

Table 2: Relative PDG measurement uncertainties compared to the expected correction order $\xi \sim 10^{-4}$.

Epistemic position

The theory makes geometric predictions that are not all currently numerically testable. This is not a deficiency of the theory but a statement about the current state of experimental physics. By analogy, many predictions of the Standard Model were theoretically consistent for decades before experimental confirmation became possible.

The testable predictions of FFGFT – CP-violation phase $\delta_{CP} = 283.3°$, weak mixing angle $\sin^2\theta_W = 3/13$, Bell correlations for massive particles – are not affected by this limitation, since they concern dimensionless ratios, not absolute mass values.

.6 Derivation of the p Exponents from Torus Topology

The regular ladder

On a 3D torus $\mathbb{T}^3$, the allowed momenta in d_{act} active dimensions are quantised. The energy of a field mode scales with the number of active dimensions:

$$E \sim \xi^{d_{\text{act}}/3} \cdot v \qquad \Rightarrow \qquad p = \frac{d_{\text{act}}}{3} \tag{31}$$

This yields the regular ladder in steps of $\Delta p = 1/3$:

p	d_{act}	Particles
$3/2$	4.5?	Electron, Up, Down
1	3	Muon, Strange
$2/3$	2	Tau, Charm
$1/3$	1	(hypothetical)
0	0	Higgs scale

Table 3: Regular p ladder: each step corresponds to one fewer active torus dimension.

Note: $p = 3/2$ formally requires $d_{\text{act}} = 4.5$ dimensions, pointing to a mixing term between the spatial and temporal components of the 4D torus. This remains an open point.

The top quark: negative exponent as inverse scale

The top quark occupies a special position:

$$p_{\text{top}} = -\frac{1}{3}, \qquad r_{\text{top}} = \frac{1}{28}, \qquad m_t \approx 172.7 \text{ GeV} \tag{32}$$

Numerical verification:

$$m_t = \frac{1}{28} \cdot \xi^{-1/3} \cdot v = \frac{1}{28} \cdot (7500)^{1/3} \cdot 246 \text{ GeV} \approx 172.0 \text{ GeV} \tag{33}$$

Geometric interpretation: All other fermions have $m_i \ll v$ – they are suppressed relative to the Higgs scale by ξ^p with $p > 0$. The top quark is the only fermion with $m_t \approx 0.7\,v$ – its Yukawa coupling is $y_t \approx 1$, meaning it couples to the Higgs field without suppression.

In torus language, $p < 0$ means the mode winds in the *inverse* direction on one torus dimension. Instead of ξ suppression, one obtains a $\xi^{-1/3}$ enhancement:

$$\xi^{-1/3} = \left(\frac{1}{\xi}\right)^{1/3} = \left(\frac{30000}{4}\right)^{1/3} = 7500^{1/3} \approx 19.57 \tag{34}$$

Geometric statement – status: ansatz

The top quark is the only fermion whose torus mode carries an **inverse winding** on one spatial dimension. This corresponds to $p = -1/3$ and yields a Yukawa coupling of order unity – marking the top as the *natural* particle of the Higgs scale, while all other fermions are suppressed.

Epistemic status: This is a geometrically motivated ansatz, not a completed derivation. The question of why exactly $d_{\text{act}} = -1$ (an inverse dimension) holds for the top remains open.

The bottom quark: transition exponent

The bottom quark has $p = 1/2$ – not an integer multiple of $1/3$:

$$p_{\text{bottom}} = \frac{1}{2}, \qquad r_{\text{bottom}} = \frac{3}{2}, \qquad m_b \approx 4.18 \text{ GeV} \tag{35}$$

Numerical verification:

$$m_b = \frac{3}{2} \cdot \xi^{1/2} \cdot v = \frac{3}{2} \cdot \sqrt{\frac{4}{30000}} \cdot 246 \text{ GeV} \approx 4.26 \text{ GeV} \tag{36}$$

$p = 1/2$ corresponds geometrically to a **2D cross-section** of the 3D torus – a surface rather than a volume or a line. This suggests that the bottom mode is localised on a 2D submanifold of the torus, physically corresponding to the transition between regular suppression and the inverse top scale.

Particle	p	**Geometry**	**Status**
Electron, Up, Down	$3/2$	3D + timecomponent	open ($d = 4.5$?)
Muon, Strange	1	3D volume	derived
Tau, Charm	$2/3$	2D surface	derived
Bottom	$1/2$	2D cross-section	ansatz
Top	$-1/3$	Inverse 1D winding	ansatz

Table 4: Geometric interpretation of the p exponents. Derived = directly from $d_{\text{act}}/3$; ansatz = motivated but not yet formally complete.

Inconsistency in Doc. 006

Doc. 006 contains at one point the assignment $p_{\text{top}} = 2/3$ (3rd generation, regular ladder). This is numerically incorrect – the correct prediction $m_t \approx 172$ GeV requires $p = -1/3$. The table in Doc. 006 carries the correct value; the body text at that location should be corrected.

.7 Harmonic Interpretation of the p Exponents

The ξ^p scale as a logarithmic resonance scale

The particle masses follow the Yukawa formula $m_i = r_i \cdot \xi^{p_i} \cdot v$. Since $\xi \ll 1$, the p axis is a **logarithmic resonance scale**:

$$\log m_i = \log r_i + p_i \cdot \log \xi + \log v \tag{37}$$

Each step $\Delta p = 1/3$ corresponds to a fixed interval on this scale:

$$\frac{m(p+1/3)}{m(p)} = \xi^{1/3} = \left(\frac{4}{30000}\right)^{1/3} \approx \frac{1}{19.6} \tag{38}$$

This is the fundamental **third** of the particle spectrum scale in FFGFT. Doc. 060 shows that the r prefactors are 5-limit ratios – the same prime factors as in just intonation in music theory.

Parallel to music theory (Doc. 060, 159)

In music theory, the overtones of a vibrating string form a harmonic series with discrete frequency ratios. The Laplace operator on a circle S^1 has eigenvalues:

$$\lambda_n = n^2, \qquad n = 0, \pm1, \pm2, \pm3, \ldots \tag{39}$$

The negative values $n < 0$ are not exceptions – they are the *other direction* of winding on the circle. Doc. 159 shows that particle masses appear as **torus overtones** with step size $\Delta p = 1/3 = 1/D$, where $D = 3$ is the spatial dimension.

The complete p scale in harmonic terms:

Particle	p	ξ^p	Harmony	Direction
Top	$-1/3$	≈ 19.6	Third upward	Inverse winding
—	0	1	Root tone	Higgs scale v
Bottom	$1/2$	$\approx 1/87$	Fourth	log. midpoint
Tau, Charm	$2/3$	$\approx 1/384$	Fifth	2 thirds
Muon, Strange	1	$\approx 1/7500$	Octave	3 thirds
e, u, d	$3/2$	$\approx 1/65 \cdot 10^4$	Fifth+Oct.	4.5 thirds

Table 5: The p exponents as harmonic intervals on the logarithmic ξ scale. The root tone is the Higgs scale v. Step size $\Delta p = 1/3$ corresponds to one third.

The negative sign as inversion – not an exception

In music theory, an interval downward (e.g. a fourth below) is just as natural as the same interval upward. The negative sign $p < 0$ means in the harmonic interpretation simply a **winding in the opposite direction** on the torus.

From Doc. 159: the Laplace operator on $\mathbb{T}^3$ has eigenvalues $\lambda_{(n_1,n_2,n_3)} = n_1^2 + n_2^2 + n_3^2$ with $n_i \in \mathbb{Z}$. Negative n_i are equivalent windings in the opposite direction.

For the top quark:

$$p_{\text{top}} = -\frac{1}{3} \quad \Longleftrightarrow \quad n_{\text{eff}} = -1 \text{ on one torus dimension} \tag{40}$$

This yields amplification rather than suppression:

$$\xi^{-1/3} = \frac{1}{\xi^{1/3}} \approx 19.6 \tag{41}$$

Why only the top? The top is the only fermion whose natural resonance frequency lies *above* the Higgs root tone v. All other fermions are suppressed by positive p values. The top winds in the opposite direction – it is the only overtone that exceeds the root tone.

The $p = 1/2$ of the bottom as logarithmic midpoint

$p = 1/2$ is in the harmonic interpretation the **geometric midpoint** between $p = 0$ (root tone) and $p = 1$ (first octave):

$$\xi^{1/2} = \sqrt{\xi} \qquad \longleftrightarrow \qquad \text{one octave lower on the log scale} \tag{42}$$

In music theory the geometric mean between two frequencies corresponds to the **tritone** – the unique self-inverting interval, equidistant on the logarithmic scale. $p = 1/2$ is therefore not an anomaly but a distinguished point of the harmonic structure.

Doc. 060 shows that the r coefficient of the bottom is $3/2$ – the **pure fifth** at 702.0 cents, the most consonant interval after the octave. Together with $p = 1/2$, the bottom quark is the harmonically simplest quark in the theory.

Summary: the complete harmonic structure

The p exponents are not arbitrary numbers. They are **winding numbers on a logarithmic torus** with step size $1/3$. Negative values are counter-windings. Half-integer values are geometric midpoints between winding levels.

This is the same structure as in music theory: overtones, octaves, fifths and fourths arise not by design, but through the **geometry of the periodic boundary condition**.

Harmonic principle

The particle spectrum scale of FFGFT is a harmonic series on the logarithmic torus. Step size $\Delta p = 1/3$ is the fundamental third. Negative p are counter-windings. $p = 1/2$ is the geometric midpoint (tritone). The top quark is the only fermion above the root tone.

Cross-references: Doc. 060 (music theory parallels, Pythagorean comma, 5-limit coefficients), Doc. 159 (harmonic series, torus overtones, vacuum catastrophe).

.8 Level 6: Spin and the Dirac Equation from Torus Topology

The central insight: winding numbers determine mass *and* spin

In the preceding sections, winding numbers (n_θ, n_ϕ) were used to derive the mass prefactors:

$$r_i = \frac{n_{\phi,i}^2}{n_{\theta,i}} \tag{43}$$

From Doc. 050, a direct connection to spin follows:

$$\text{Spin-}s \longleftrightarrow \frac{n_\phi}{n_\theta} = 2s \tag{44}$$

This immediately gives:

$$\boxed{r_i = \frac{n_{\phi,i}^2}{n_{\theta,i}} = n_{\phi,i} \cdot \underbrace{\frac{n_{\phi,i}}{n_{\theta,i}}}_{=2s_i} = 2s_i \cdot n_{\phi,i}} \tag{45}$$

Important clarification: The spin-$1/2$ condition $n_\phi/n_\theta = 1$ applies to the *spinor component* of the winding, not to the mass-generating component. Mass and spin arise from *different projections* of the same winding numbers onto the 4D torus. This is detailed in Doc. 051.

Clifford algebra from torus geometry

The fundamental Dirac equation in FFGFT reads (Doc. 050):

$$(i\,\mathbf{e}_\mu^{(\text{frac})}\,\partial^\mu - m)\,\Psi = 0 \tag{46}$$

with the fractal Clifford algebra:

$$\mathbf{e}_\mu^{(\text{frac})}\mathbf{e}_\nu^{(\text{frac})} + \mathbf{e}_\nu^{(\text{frac})}\mathbf{e}_\mu^{(\text{frac})} = 2\,g_{\mu\nu}^{(\text{frac})} \tag{47}$$

The 4×4 Dirac matrices γ^μ are *one* matrix representation of this abstract structure – not the physics itself. The physics is the torus geometry.

Spin-$1/2$ as a topological property

The well-known 720° property of spin-$1/2$:

$$R(360^\circ)\,\Psi = -\Psi \qquad R(720^\circ)\,\Psi = +\Psi \tag{48}$$

follows directly from the torus winding (Doc. 050):

Spin as topology

Spin-$1/2$ corresponds to winding $(n_\theta, n_\phi) = (1,1)$ on the torus. One 360° rotation = one traversal of the winding → sign change. One 720° rotation = two traversals → return to identity. This is **pure topology**, not a mysterious quantum property.

The T0-FFGFT Dirac equation with dynamic mass

In FFGFT, mass is not a constant but a dynamic field (from the fundamental relation $\tilde{T} \cdot m = 1$). The Dirac equation becomes:

$$\left(i\,\mathbf{e}_\mu^{(\text{frac})}\,\partial^\mu - m(x,t)\right)\Psi = 0 \tag{49}$$

with $m(x,t) = m_0 + \delta m(x,t)$. For small excitations:

$$\left(i\,\mathbf{e}_\mu^{(\text{frac})}\,\partial^\mu - m_0\right)\Psi = \delta m(x,t) \cdot \Psi \tag{50}$$

The right-hand term couples the Dirac spinor Ψ to the mass excitation field δm. The corresponding Lagrangian term is:

$$\mathscr{L}_{\text{Dirac-T0-FFGFT}} = \bar{\Psi}\left(i\,\mathbf{e}_\mu^{(\text{frac})}\,\partial^\mu - m_0\right)\Psi - \delta m \cdot \bar{\Psi}\Psi \tag{51}$$

Connection to the Lagrangian hierarchy

The complete FFGFT Lagrangian density unifies all levels:

$$\boxed{\mathscr{L}_{\text{FFGFT}} = \underbrace{\varepsilon(\partial\delta m)^2}_{\text{Level 2}} + \underbrace{\Delta\mathscr{L}_{\text{Torus}}}_{\text{Level 3}} + \underbrace{\bar{\Psi}(i\,\mathbf{e}_{\mu}^{(\text{frac})}\,\partial^{\mu} - m_0)\Psi - \delta m\,\bar{\Psi}\Psi}_{\text{Level 6}}} \tag{52}$$

The masses $m_0 = m_i = r_i \cdot \xi^{p_i} \cdot v$ follow from level 4 (winding numbers). The torus correction term $\Delta\mathscr{L}$ from level 3 modifies the metric $g_{\mu\nu}^{(\text{frac})}$. Everything depends on ξ.

Open point: quarks and colour charge

For quarks (SU(3) structure, colour charge), the spinor structure is more complex. The winding numbers must be extended by a colour quantum number $n_c = 3$. This is outlined conceptually in Doc. 051 but not yet fully formalised.
Cross-references: Doc. 050 (Dirac equation geometrically, simplified), Doc. 051 (complete technical treatment).

.9 Lagrangian Extension: Precision Beyond the SM

Starting point and goal

The Standard Model treats the Yukawa couplings y_i as free parameters – 12 independent numbers without geometric origin. The T0-FFGFT Lagrangian extension has a concrete goal: to **derive** the r_i and p_i from torus geometry, not merely describe them. This enables calculations that the SM structurally cannot perform.

The extended Lagrangian with resonance modes

The complete FFGFT Lagrangian for a fermion field with winding numbers (n_θ, n_ϕ) on the 4D torus reads:

$$\mathscr{L}_{(n_\theta,n_\phi)} = \varepsilon_{(n)} \cdot (\partial_\mu \delta m)^2 + \frac{\kappa}{L_0^2} \cdot \frac{n_\phi^2}{n_\theta} \cdot \xi^{d_{\text{act}}/3} \cdot \delta m^2 + \Delta\mathscr{L}_{\text{Torus}} \tag{53}$$

The second term is the **resonance mass term**:

- $n_\phi^2/n_\theta = r_i$ – the Yukawa prefactor follows geometrically from the winding numbers
- $\xi^{d_{\text{act}}/3} = \xi^{p_i}$ – the exponent follows from the number of active torus dimensions
- $\kappa = \varepsilon \cdot v^2$ – coupling strength to the Higgs scale

From the Euler-Lagrange equation for this term, the effective mass of the mode follows:

$$m_i^2 = \frac{\kappa}{L_0^2} \cdot \frac{n_{\phi,i}^2}{n_{\theta,i}} \cdot \xi^{d_{\text{act},i}/3} = r_i \cdot \xi^{p_i} \cdot v^2 \cdot \frac{1}{L_0^2} \tag{54}$$

This is exactly the Yukawa formula $m_i = r_i \cdot \xi^{p_i} \cdot v$ – now derived from the Lagrangian, not postulated.

What this Lagrangian achieves

What the SM cannot do, T0-FFGFT can:

Quantity	SM	T0-FFGFT
Yukawa couplings	12 free parameters	$r_i = n_\phi^2/n_\theta$ from winding
Mass exponents	No structure	$p_i = d_{\text{act}}/3$ from topology
Generation number	Empirically 3	Torus dimensions: $d = 3$
Neutrino masses	Seesaw ad hoc	ξ^2 suppression geometric
Koide relation	Accidental	Consequence of p_i structure

Table 6: Improvements from the extended T0-FFGFT Lagrangian compared to the Standard Model.

Higher order corrections

The correction term $\Delta\mathscr{L}_{\text{Torus}}$ from Section 3 generates mode-specific corrections. Angle-averaged:

$$m_i^{(\text{eff})} = m_i^{(0)} \cdot \left(1 + \sum_{k=1}^{N} c_k \cdot \xi^k \cdot f(p_i, k)\right) \tag{55}$$

where $f(p_i, k)$ is the generation-dependent weighting of the k-th correction term. These corrections go beyond the SM because they encode the torus curvature mode by mode.

Why better than SM:

1. **Structural**: The Yukawa couplings are not free – they are fixed by (n_θ, n_ϕ). The SM has no predictive power here.
2. **Precision**: The corrections $c_k \cdot \xi^k$ are geometrically determined, not regularised by renormalisation of divergences. The SM correction scheme (QED up to α^5) is model-dependent.
3. **Consistency**: Mass, spin and generation follow from the same winding numbers – in the SM these are separate degrees of freedom.

Open question: own correction formalism or SM copy?

The obvious next step would be to calculate radiative corrections analogously to the SM:

$$m_i^{(\text{T0-phys})} = m_i^{(0)} \cdot \left(1 + \frac{\alpha_{\text{T0}}}{2\pi} \cdot g(p_i) + \ldots\right) \tag{56}$$

with $\alpha_{\text{T0}} = \xi \cdot E_0^2 = 1$ in T0 units.

But caution is warranted: if one adopts the same formalism as the SM – Feynman diagrams, loop integrals, renormalisation scheme – that would not be new physics, but the SM with geometric regularisation. This explains similar numbers, but not why they *must* be geometric.

What the torus does structurally differently:

The 4D torus has **discrete momentum modes** – no continuous momentum integration. The consequences:

- No UV divergences – the torus provides a natural cutoff at $1/L_0$, without regularisation
- Loop integrals become **finite sums** over winding numbers (n_θ, n_ϕ)
- Renormalisation would not be needed – or would have a fundamentally different geometric meaning

The deeper possibility:

The corrections in T0-FFGFT may follow directly from the recursion formula:

$$\Psi(t) = G(\Psi(t - T_{\text{rev}}), \xi) \tag{57}$$

The recursion *is* the quantum correction – not an addition to it. Each recursion depth k corresponds to a correction order, and c_k is not a renormalisation parameter but a geometric coefficient from G.

Open fundamental question

Do the T0-FFGFT corrections follow from the recursion structure $\Psi(t) = G(\Psi(t - T_{\text{rev}}), \xi)$ without SM formalism – or is an SM-analogous correction scheme needed?

The answer determines whether T0-FFGFT delivers **structurally new physics** or a geometrically grounded variant of the SM.

Cross-references: Section 3 (torus curvature correction), Section 4 (winding numbers), Doc. 183, 184 (recursion formula).

.10 Planck Units as Translation Factors

Two different roles of Planck quantities

In the Standard Model and conventional quantum gravity, Planck units are treated as **fundamental constants**: $G = \hbar = c = 1$ (Planck units). The formulas appear simplified, but the constants remain conceptually fundamental.

In FFGFT, the role is different: Planck quantities are **translation factors** between the ratio-based T0-FFGFT unit system and the SI measurement system. They are themselves derivable from ξ (Doc. 013):

$$G = \frac{\xi^2}{4\,m_e} \quad \text{(gravitational constant from } \xi\text{)} \tag{58}$$

$$\ell_P = \sqrt{G} = \frac{\xi}{2\sqrt{m_e}} \quad \text{(Planck length from } \xi\text{)} \tag{59}$$

$$\alpha = \xi \cdot E_0^2 = 1 \quad \text{(in T0-FFGFT units; translation to SI: } 1/137.036\text{)} \tag{60}$$

What SM formulas actually encode

The formulas of the SM containing G, $\hbar$ and c do not encode fundamental constants – they encode **translation ratios** between the dimensionless geometric T0-FFGFT system and the SI measurement system:

SM formula	SM interpretation	T0-FFGFT interpretation
$\ell_P = \sqrt{\hbar G/c^3}$	Definition, fundamental	$\xi/(2\sqrt{m_e})$, from ξ
$G = 6.674 \times 10^{-11}$	Free constant	$\xi^2/(4m_e)$, geometric
$\alpha = 1/137$	Universal constant	Unit convention
$c = 1$ (Planck)	Convention	Ratio ℓ_P/t_P

Table 7: Planck quantities in SM (as constants) vs. T0-FFGFT (as translation factors between unit systems). Doc. 013, 014, 015.

Ratio-based natural units

In T0-FFGFT natural units there are no arbitrarily set constants. Instead, all quantities are **ratios**:

$$\frac{m_i}{m_e} = r_i \cdot \xi^{p_i} \cdot \frac{v}{m_e} \qquad \text{(pure ratio)} \tag{61}$$

The Planck length ℓ_P serves as a bridge between the geometric scale $L_0 = \xi \cdot \ell_P$ and the SI measurement in metres. It is not a foundation of the theory but a calibration point.

Consequence for the epistemic barrier: The physics community has interpreted Planck units as fundamental constants and searched for why G, $\hbar$, c have exactly these values. In T0-FFGFT that is the wrong question – the right questions are: Which geometric ratios produce these measured values? Answer: $\xi = 4/30000$.
Cross-references: Doc. 012 (G from ξ), Doc. 013 (SI translation), Doc. 014 (natural units), Doc. 015 (unit systematics).

.11 Epistemic Barriers: Why the Community Did Not Reach the Same Conclusion Earlier

The misinterpretation of the fine structure constant

The fine structure constant $\alpha \approx 1/137.036$ is treated in the Standard Model as a universally fixed dimensionless parameter – independent of the chosen unit system. This is a **misinterpretation**.

From Doc. 011, 013, 014 and 015 it follows clearly:

Unit system	α	Interpretation
SI	$1/137.036$	Measured value, system-dependent
T0-FFGFT units	1 (exact)	Geometric identity
Gaussian units	1 (exact)	Different convention

Table 8: α is not a universal constant but a translation convention between unit systems (Doc. 013, 015).

The numerical value $1/137.036$ does not encode a fundamental property of nature, but the **ratio of two units of measure** in the SI system. In T0-FFGFT natural units:

$$\alpha_{\text{T0-FFGFT}} = \xi \cdot E_0^2 = 1 \qquad \text{(geometric identity, Doc. 011)} \tag{62}$$

The search for an "explanation" of $1/137$ was therefore from the outset a search for the shadow of a unit system – not for the geometry behind it. Feynman called α the greatest mystery of physics. It is no mystery once the correct unit system is chosen.

The enforced separation of QM and SM

The second epistemic barrier is the notion that quantum mechanics (QM) and the Standard Model (SM) must not be mixed – they operate at different scales, with different formalisms, and the boundaries are historically entrenched.

This separation prevented the geometric structure common to both from becoming visible:

Phenomenon	QM/SM treats it as	FFGFT shows
Fermion masses	Yukawa parameters (free)	$r_i \cdot \xi^{p_i} \cdot v$, geometric
α	Universal constant	Unit convention, $\alpha_{\text{T0-FFGFT}} = 1$
Spin-$1/2$	Matrix property	Torus winding number
Neutrino mass	Seesaw mechanism	$\xi^2 \cdot m_e$, topological
Vacuum energy	Renormalisation needed	Torus topology converges

Table 9: Phenomena treated by SM and QM as separate problems, but which follow in FFGFT from the same torus geometry.

Geometric gravity

A third point concerns the gravitational constant G. In the SM, G is a free parameter without geometric derivation. In FFGFT, G follows from ξ and the torus geometry – shown in the corresponding documents of the series. The identification $G \sim \xi \cdot \ell_P^2/\hbar$ connects gravity directly to the single free parameter.

Summary of epistemic barriers

Three barriers prevented the community from reaching the same conclusion earlier:

1. α **as a universal fixed value** – rather than as a unit convention. The geometric derivation $\alpha = \xi \cdot E_0^2$ is invisible within the SI framework.
2. **Enforced SM/QM separation** – prevents recognition of the common torus geometry. Fermion masses, spin and mixing angles follow from the same source.
3. **Renormalisation as a solution** – rather than a symptom. The vacuum catastrophe arises only in the wrong (non-toroidal) background. In the correct topology, the harmonic series converges (Doc. 159).

Cross-references: Doc. 011 (α derivation), Doc. 012 (gravitational constant), Doc. 013 (SI vs. T0-FFGFT units), Doc. 014 (natural units), Doc. 015 (unit systematics), Doc. 159 (vacuum catastrophe and torus).

.12 Open Question: Are All 12 Fermions Fundamental Torus Resonances?

The Euler Tonnetz and the limits of harmony

In classical music theory, the prime numbers 2, 3 and 5 form the three axes of the **Euler Tonnetz** – the complete lattice of all consonant intervals in just intonation (5-limit).

The prime number 7 produces the **natural seventh** – an interval that cannot be resolved in any pure tuning system. It lies literally outside harmonic space: a fourth dimension would be needed to embed it. The prime number 13 has no role whatsoever in Western harmony – it lies beyond any known consonant structure.

5-limit analysis of the fermion prefactors

The r prefactors of the FFGFT fermions were systematically analysed for their prime factor structure in Doc. 060:

Particle	r	Prime factors	5-limit?	Musical interval
Electron	$4/3$	$2^2/3$	✓	Pure fourth
Muon	$16/5$	$2^4/5$	✓	Minor sixth + octave
Tau	$25/9$	$5^2/3^2$	✓	Augmented fourth
Up	6	$2 \cdot 3$	✓	Fifth + 2 octaves
Down	$25/2$	$5^2/2$	✓	2 major thirds
Charm	2	2	✓	Octave
Bottom	$3/2$	$3/2$	✓	Pure fifth
Strange	$26/9$	$2 \cdot \mathbf{13}/3^2$	$\times$	outside (13)
Top	$1/28$	$1/(2^2 \cdot \mathbf{7})$	$\times$	natural seventh (7)

Table 10: 5-limit analysis of the fermion prefactors. Seven of nine fermions are consonant 5-limit ratios. Strange (factor 13) and Top (factor 7) lie outside the harmonic lattice.

Geometric consequence

In the Euler Tonnetz, 2, 3 and 5 each require one axis – a 3-dimensional lattice. The prime 7 would require a **fourth dimension**, 13 a fifth. In the FFGFT torus geometry, the 4D torus has exactly 4 dimensions – the fourth being the time dimension.

Possible interpretation:

- The 7 fermions with 5-limit prefactors are **genuine spatial resonances** on $\mathbb{T}^3$
- Top ($\times 7$) may be a resonance involving the **time dimension** – which would explain why its mass lies close to the Higgs scale v
- Strange ($\times 13$) lies **outside any known harmonic structure** – strongest candidate for a non-fundamental resonance

Logarithmic combination analysis

In a harmonic resonance scale, adding exponents means multiplying masses:

$$m_{\text{comb}} = r_1 \cdot r_2 \cdot \xi^{p_1+p_2} \cdot v \quad \text{(logarithmic combination)} \tag{63}$$

A numerical check of all pairs yields two notable matches:

Combination	p_{sum}	r_{prod}	m_{comb}	Comparison
Charm ⊗ Strange	$5/3$	$52/9$	495 keV	m_e: T0 Δ=2.1%; PDG Δ=3.1%
Top ⊗ Tau	$1/3$	$25/252$	1.25 GeV	m_c: T0 Δ=2.9%; PDG Δ=1.7%

Table 11: Two pairs whose combination mass lies close to a known particle mass. However the p exponents do not exactly match the ladder ($5/3 \neq 3/2$, $1/3 \notin$ ladder).

The matches are interesting but not conclusive: the p exponents of the combinations do not fall exactly on the known ladder positions.

The Koide formula as the same phenomenon

The Koide formula (1981) connects the lepton masses:

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \quad \text{(experimental accuracy} < 0.001\%\text{)} \tag{64}$$

Doc. 116 shows: the Koide formula is not an independent empirical relation but follows directly from the T0-FFGFT structure. The reason lies in the **same phenomenon**: the $\sqrt{m}$ terms contain $p/2$ exponents:

$$\sqrt{m_i} = \sqrt{r_i} \cdot \xi^{p_i/2} \cdot \sqrt{v} \tag{65}$$

Lepton	p	$p/2$	$\sqrt{r_i}$
Electron	$3/2$	$3/4$	$\sqrt{4/3} \approx 1.155$
Muon	1	$1/2$	$\sqrt{16/5} \approx 1.789$
Tau	$2/3$	$1/3$	$\sqrt{25/9} \approx 1.667$

Table 12: The $\sqrt{m}$ terms of the Koide formula generate a half-ladder with $p/2$ exponents: $3/4$, $1/2$, $1/3$ – step size $\Delta(p/2) = 1/4$ or $1/6$.

The T0-FFGFT result: with **T0-FFGFT ideal values**, $Q = 0.6677$ – deviation 0.16%. With **PDG measured values**: $Q = 0.66666$ – deviation 0.0009%.

At first glance this looks like a deficiency of the ideal values. A closer analysis shows otherwise:

PDG measured values are model-dependent (script `messwert_analyse.py`): The PDG lepton masses are determined using SM radiative corrections (QED up to order α^5, electroweak corrections). These corrections are ~ 0.3–0.5% for m_τ. The T0-FFGFT deviation for m_τ is $+0.46\%$ – **the same order of magnitude**.

The σ deviations (m_e: $3.7 \times 10^{14}\,\sigma$, m_μ: $2.6 \times 10^5\,\sigma$) are physically meaningless – the statistical measurement uncertainty of PDG values is irrelevant as long as the systematic model dependence (~ 0.3–1%) is larger than the T0 deviation.

Consequence: The T0-FFGFT calculations may be better than the direct comparison with PDG values suggests. The remaining 0.5–1% deviations lie in the regime where SM radiative corrections and T0-FFGFT corrections diverge. Without a complete T0 scheme for radiative corrections, it cannot be decided whether the deviations originate from:

1. $\xi = 4/30000$ as a slight rational approximation,
2. $v = 246.22$ GeV as an SM-dependent measured value, or
3. the difference between SM and T0-FFGFT radiative corrections

or a combination of all three.

The geometric statement of T0-FFGFT remains precise: the p values $\{3/2, 1, 2/3\}$ explain **why** Q lies close to $2/3$. The fractions r_i and exponents p_i are geometrically exact – the residual deviations are not errors of the geometry.

Numerically verified (scripts `koide_test.py`, `koide_kfrak_test.py`, `koide_e0_test.py`, `koide_korrekt.py`): Q is not a pure rational ratio formula. It contains $\xi^{1/2}$ and $\xi^{1/6}$ – both transcendental. v cancels, ξ does not. The products $r_i \cdot \xi^{p_i}$ must be treated as indivisible units (unit error if cancelled separately).

Cross-references: Doc. 116 (Koide formula and T0-FFGFT), Doc. 060 (5-limit analysis).

Epistemic status

Open question – not part of the completed T0-FFGFT

Strange and Top are the only fermions whose r prefactors contain the primes 13 and 7 respectively – outside the 5-limit scheme of all other fermions. By analogy with the Euler Tonnetz, this suggests that these two quarks may not be independent fundamental resonances of the 3D torus. A complete derivation is absent. This is an open question for the further development of the theory.

Cross-reference: Doc. 060 (5-limit analysis, Euler Tonnetz, music theory parallels).

.13 Concluding Summary: Structural Comparison with the SM

What the Standard Model says about fermion masses

The SM describes 9 charged fermion masses through 12 free Yukawa parameters:

$$m_i = y_i \cdot \frac{v}{\sqrt{2}} \tag{66}$$

This is not an explanation – it is a renaming. The 12 numbers y_i could take any value whatsoever; the SM would work equally well with entirely different fermion masses. It does not answer:

- Why are there exactly three generations?
- Why is $m_\mu/m_e \approx 207$?
- Why do the leptons satisfy the Koide formula?
- Why is $m_t \approx v$ while all other fermions lie below 5 GeV?

To all these questions the SM answers: "Those are just the measured values."

What T0-FFGFT achieves with one parameter

T0-FFGFT explains the same masses from $\xi = 4/30000$:

Achievement	SM	T0-FFGFT
Free mass parameters	12 Yukawa couplings	**1** (ξ)
r_i prefactors	Arbitrary	n_ϕ^2/n_θ – winding
p_i exponents	No concept	$d_{\text{act}}/3$ – topology
Generation number	3 (empirical)	3 (dim. $\mathbb{T}^3$)
$m_t \approx v$	Accident ($y_t \approx 1$)	$p = -1/3$ – inv. winding
Koide $Q = 2/3$	Pure coincidence	From p_i structure
Neutrino masses	Seesaw (ad hoc)	$\xi^2 \cdot m_e$ – geometric
5-limit structure	None	7/9 fermions consonant

Table 13: Structural comparison SM vs. T0-FFGFT.

The honest limitation

Important limitation

T0-FFGFT currently provides **no numerically better calculations** than the SM. The ideal values deviate 0.5–1% from PDG measured values – the SM matches the same values by direct parametrisation by definition.
The superiority of T0-FFGFT is **structural, not numerical**: it reduces 12 free parameters to 1 and provides structural explanations where the SM merely enters measured values.

The remaining 0.84% deviation lies in the same order of magnitude as the SM radiative corrections in the PDG values (~ 0.3–0.5%). Whether the deviation is a deficiency of T0-FFGFT or an artefact of the SM model dependence of the measured values cannot be decided without a T0-native correction scheme.

Conclusion

Structural finding

The SM **describes** fermion masses with 12 parameters.
T0-FFGFT **explains** the same masses with 1 parameter.

This is a categorially different kind of theory – regardless of whether all details of the derivation are yet complete.

Furthermore, T0-FFGFT achieves something neither QM nor GR alone can: the fundamental relation $\tilde{T} \cdot m = 1$ unites both theories in a single geometric framework. Quantum mechanics describes wave functions on a fixed spacetime background. General relativity describes spacetime curvature without quantum structure. T0-FFGFT encodes both in the torus geometry: the mass field $m(x,t)$ is simultaneously the source of spacetime curvature (GR) and the carrier of quantum resonances (QM).

What quantum modes really are: The discrete modes of QM are the same as the torus modes – not an analogy, but an identity. Winding numbers $(n_\theta, n_\phi) \in \mathbb{Z}$ are integers because a field on a compact torus can only wind an integer number of times. This is topological necessity, not an imposed postulate. T0-FFGFT does **not quantise** – it shows that quantisation is a topological consequence.

Even if T0-FFGFT should prove incomplete, it has already shown: the fermion masses are not arbitrary. They follow a rational torus resonance structure with 5-limit prime factors. The SM has had only one answer to this for 50 years: "Those are just the measured values."

List of Symbols

Symbol	Meaning
ξ	Fundamental geometric parameter of T0-FFGFT, $\xi = 4/30000 \approx 1.333 \times 10^{-4}$. Single free parameter; determines all constants of nature.
$\tilde{T}(x,t)$	Dynamic time field (reciprocal mass field); fundamental relation: $\tilde{T} \cdot m = 1$.
$\delta m(x,t)$	Mass excitation field; small deviation from background value m_0: $m = m_0 + \delta m$.
$\mathscr{L}_0$	Free Lagrangian density: $\mathscr{L}_0 = \varepsilon \cdot (\partial \delta m)^2$.
$\Delta\mathscr{L}_{\text{Torus}}$	Torus curvature correction to the free Lagrangian density; first order in ξ.
ε	Coupling constant of the free Lagrangian; $\varepsilon_i = \xi \cdot m_i^2$.
r_i	Rational Yukawa prefactor of particle i; derived from winding numbers: $r_i = n_{\phi,i}^2 / n_{\theta,i}$. 5-limit ratio (Doc. 060).
p_i	Torus exponent of particle i; determines the ξ suppression: $m_i = r_i \cdot \xi^{p_i} \cdot v$. Step size $\Delta p = 1/3$.

Symbol	Meaning
v	Higgs vacuum expectation value (VEV), $v \approx 246.22$ GeV; fundamental scale of all fermion masses.
n_θ, n_ϕ	Winding numbers on the torus; determine mass ($r_i = n_\phi^2/n_\theta$) and spin ($n_\phi/n_\theta = 2s$).
s	Spin of a particle; topological property of the torus winding.
n_c	Colour quantum number of quarks; $n_c = 3$ (SU(3) structure).
D_f	Fractal spatial dimension; $D_f^{\text{space}} = 3 - \xi \approx 2.9999$ (derived in Doc. 009, 133).
$\mathbf{e}_\mu^{(\text{frac})}$	Fractal Clifford algebra basis vectors; anticommutator relation: $\mathbf{e}_\mu\mathbf{e}_\nu + \mathbf{e}_\nu\mathbf{e}_\mu = 2g_{\mu\nu}^{(\text{frac})}$.
$\Psi, \bar{\Psi}$	Dirac spinor and adjoint spinor; topological object in the spin bundle over the torus.
ℓ_P	Planck length; in T0-FFGFT not a fundamental parameter but a translation factor: $\ell_P = \xi/(2\sqrt{m_e})$ (Doc. 013).
L_0	Fundamental T0-FFGFT length; $L_0 = \xi \cdot \ell_P \approx 2.155\times10^{-39}$ m.
E_0	Characteristic energy scale; $E_0 = \sqrt{m_e \cdot m_\mu} \approx 7.398$ MeV.
α	Fine structure constant; in T0-FFGFT units $\alpha = 1$ (exact), in SI units $\alpha = 1/137.036$. Unit convention, not a universal fixed value (Doc. 011).
K_{frak}	Fractal correction; $K_{\text{frak}} = 1-100\xi \approx 0.9867$. Arises from cumulative scale flow over 17 orders of magnitude (Doc. 133).
λ_h	Higgs self-coupling, $\lambda_h \approx 0.129$.
m_h	Higgs mass, $m_h \approx 125.10$ GeV.
G	Gravitational constant; in T0-FFGFT not a free parameter: $G = \xi^2/(4m_e)$ (Doc. 012).
$\Psi(t)$	FFGFT recursion field; $\Psi(t) = G(\Psi(t-T_{\text{rev}}), \xi)$ (Doc. 183, 184).
T_{rev}	Recursion period in the FFGFT time field (Doc. 184).

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